

National Quant Finance Olympiad 2026

Implied Volatility Surface Reconstruction — Methodology

1. Problem Framing

The task is to reconstruct missing implied volatilities (IVs) on a NIFTY50-style option surface. With 30–45% of observations missing, this is a **spatial interpolation and regression problem** with strong financial structure constraints. The evaluation metric is RMSE on held-out ground-truth IVs.

The key insight is that the IV surface is *not arbitrary* — it has well-known mathematical shapes (the volatility smile and term structure), and it must satisfy no-arbitrage constraints. Any model that ignores this structure leaves significant accuracy on the table.

2. Architecture: Four-Layer Stacking Pipeline

We use a **four-layer stacking approach** that combines quant finance theory with modern machine learning. Each layer corrects the weaknesses of the previous one.

Layer	Method	Purpose
1	SVI Surface Calibration	Quant anchor — financially correct baseline
2	Regime Detection (K-Means)	Market context for each date
3	HistGradientBoosting + ExtraTrees + Ridge	ML corrects where SVI is wrong
4	Arbitrage-Free Post-Processing	Calendar monotonicity, put-call parity

3. Layer 1 — SVI Surface Calibration (Quant Anchor)

The **Stochastic Volatility Inspired (SVI)** model is the industry standard for parameterising a single maturity slice of the IV surface. For each (date, maturity) pair, we fit:

$$w(k) = a + b * (rho*(k-m) + sqrt((k-m)^2 + sigma^2))$$

where $w = IV^2 * \tau$ is total implied variance and $k = \log(K/S)$ is log-moneyness. The five parameters $\{a, b, \rho, m, \sigma\}$ are calibrated by minimizing mean-squared error on observed IVs using L-BFGS-B with multiple starting points to avoid local minima. We achieved **520 successful SVI fits** across all date/maturity combinations.

Why SVI over polynomial fit? SVI extrapolates correctly to extreme strikes (the wings), is arbitrage-free by construction when the Lee moment formula is satisfied, and is used by major banks and hedge funds in production. A polynomial can become negative or explode in the wings.

4. Layer 2 — Regime Detection

The problem statement hints at three hidden regimes. We identify them using **K-Means clustering (k=3)** on two date-level signals:

- ATM 1M IV level: how stressed is the market today?
- Term structure slope: 6M ATM IV minus 1M ATM IV (inverted = stress)

Regime	ATM 1M IV	Slope (6M-1M)	Dates	Interpretation
Calm	~17.6%	~-1.6%	46	Low vol, flat/normal term structure
Normal	~19.9%	~-2.4%	54	Baseline conditions
Turbulent	~27.2%	~-3.0%	30	Stress: elevated IV, steep inversion

The regime label is passed as a feature to the ML models. This dramatically improves predictions because the smile shape and term structure behave very differently across regimes.

5. Layer 3 — Machine Learning Ensemble

5.1 Feature Engineering (40 features)

We engineer features across five groups:

Group	Features
Option structure	moneyness, $\log(K/S)$, $\log(K/S)^2$, $ \log(K/S) $, τ , $\log(\tau)$, $1/\tau$, $\sqrt{\tau}$, call/put flags, maturity OHE
Interactions	$\log(K/S)*\tau$, $\log(K/S)^2*\tau$, $ \log(K/S) *\tau$, OTM flag, OTM distance
Temporal	sin/cos encoding of day-of-week, week-of-year, month (prevents leakage from raw date integers)
Surface context	ATM IV by maturity, term structure slope, left skew, IV standard deviation, regime label
SVI anchor	SVI model prediction for this (date, maturity, moneyness), distance to SVI ATM centre

5.2 Models

Model A — HistGradientBoosting (sklearn's LightGBM equivalent): 800 rounds, `max_leaf_nodes=63`, `learning_rate=0.03`. Captures non-linear interactions. Trained with early stopping on 10% holdout.

Model B — ExtraTrees: 500 trees, `max_depth=20`. Uses fully random splits for a different bias/variance tradeoff, making it complementary to gradient boosting in the blend.

Model C — Ridge Regression on polynomial features: linear model that provides stable extrapolation at the tails where tree models can behave erratically.

5.3 Time-Ordered Cross-Validation

We strictly split by date: first 80% of trading dates for training, last 20% for validation. This prevents data leakage — we never use future information to predict past IVs. This is the correct protocol for financial time series.

5.4 Optimal Blend

We minimize validation RMSE over a weighted combination of all four predictors (Models A, B, C and SVI) using a grid search followed by Nelder-Mead refinement. The optimal weights were approximately: HistGBM 72%, ExtraTrees 17%, Ridge 11%, SVI 0%. The SVI weight being near zero indicates the ML models learned the surface structure well enough to subsume the parametric fit — but the SVI prediction is still invaluable as a *feature*.

6. Layer 4 — Arbitrage-Free Post-Processing

We enforce three financial constraints as a post-processing step:

Put-Call Parity: Black-Scholes implies that call and put IVs must be equal for the same (date, strike, maturity). We enforce this by averaging our predictions for each call/put pair that appears in the unknown set. This corrected 364 pairs.

Calendar Spread Monotonicity: Total variance $w(T) = IV^2 * T$ must be non-decreasing in T for a fixed strike. We use **Isotonic Regression** (Pool Adjacent Violators algorithm) to enforce this monotonicity constraint while minimizing perturbation to our predictions.

Hard Bounds: All predictions are clipped to [5.0, 80.0] per competition rules. Negative IVs are economically invalid. No predictions violated these bounds after the above corrections.

7. Validation Results

Model	Val RMSE	Notes
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SVI only (quant baseline)	0.714	Parametric surface fit only
Ridge regression	0.620	Linear model with polynomial features
ExtraTrees	0.635	Ensemble of randomized trees
HistGradientBoosting (LightGBM-equiv)	0.565	Best single model
Optimally blended ensemble	0.560	Final submission score (estimated)

8. Why This Beats the Suggested Approaches

The problem sheet suggests Options A–D independently. Our solution goes beyond by **stacking** all four into a single pipeline where each component reinforces the others:

- SVI alone (Option A) achieves RMSE ~ 0.71 — good quant baseline but misses cross-date patterns
- ML alone (Option B) achieves ~ 0.59 — but can violate arbitrage and misses financial structure
- Using SVI as an ML *feature* (not just as a prediction) is the key innovation — the ML model learns when to trust and when to deviate from the SVI surface
- Regime detection (Option C) provides the 'context' both SVI and ML need to be accurate
- Arbitrage constraints (Option D) add the 10% bonus score component and improve surface quality

9. Reproducibility

The full pipeline is in **solution.py**. Run with:

```
pip install -r requirements.txt
python solution.py
```

All random seeds are fixed (random_state=42). The pipeline is deterministic and produces identical submission.csv on each run. Runtime: approximately 3–5 minutes on a standard laptop.